Rotaract Club of Xiamen University Malaysia

Rotary Club of Kuala Lumpur Dynamics

Rotary Club of Tun Razak Exchange

SDC Hackathon

Phase 1 : Preliminary Project Proposal

Hosted by Xiamen University Malaysia (XMUM) AI Club

Submission Deadline: 25 October 2026

Instructions: Complete every field below and submit as a single PDF document named “SDC2026_TeamName_Preliminary.pdf” before the deadline above. Answers must comply with the word limits stated for each open-ended question. Incomplete submissions may not be eligible for review.

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1. TEAM AND PROJECT OVERVIEW

Team Name

Team X

Selected Domain

Domain 2

Selected SDG Track (1–17)

SDG 8&SDG12

Project / Solution Name

OrderWeave

Name (Leader)

Liu Shiwei

Email & Phone (Leader)

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University Affiliation(s)

Xiamen University Malaysia

Number of Team Members

4

2. EXECUTIVE SUMMARY & PROBLEM STATEMENT

2.1. The Problem Context

Describe the specific, real-world challenge your team is tackling. Who is affected by this problem, and why does it matter? Highlight the critical social, environmental, or economic issues involved.

[Maximum 250 words]

Small food vendors near university campuses commonly collect student orders through WeChat groups. Order messages are unstructured, informal and mixed with casual conversations, containing abbreviations, additional requests and ambiguous remarks. Vendors manually sort orders, count quantities and prepare delivery lists every day. This repetitive manual work is time-consuming and error-prone, which may lead to missing orders , wrong meals and food waste.

The affected groups are these micro-entrepreneurs and student customers. For vendors, extra labour reduces operational efficiency and income stability (aligned with SDG 8: Decent Work and Economic Growth). Miscalculated order quantities cause over-preparation of food, creating unnecessary food waste (aligned with SDG 12: Responsible Consumption and Production). Existing food delivery platforms are too costly and complex for these small-scale merchants. There lacks a lightweight tool to organise messy group messages into structured orders while keeping merchants in full control for final approval.

2. EXECUTIVE SUMMARY & PROBLEM STATEMENT

2.2. The Proposed AI Solution

Provide a high-level overview of your proposed application. How does it apply artificial intelligence to solve the problem stated above? Briefly mention the tools or technical skills gained from the SDC Phase 1 workshops (e.g., GenAI APIs, database architecture, cloud hosting) that you intend to leverage.

[Maximum 300 words]

Our proposed lightweight AI system converts unstructured WeChat group food-ordering text messages into structured order records for campus micro-vendors.
We utilise generative AI APIs to parse informal, colloquial student messages, extracting key fields including customer name, food item, quantity, special remarks and delivery location. The AI is designed to flag ambiguous or incomplete messages instead of fabricating information, preventing hallucination errors.
After AI parsing, all parsed orders are gathered in a dashboard for merchant review and approval. Unapproved entries will not flow into the final cooking and delivery list. Approved orders are automatically aggregated into a food preparation summary and delivery checklist, with operation logs stored.
The system supports flexible configuration for different vendors, including menu setup, order cut-off time and delivery rules. We will use database architecture to store menus and order records, with cloud hosting for deployment. A simple menu display page is built as an optional future extension. Human-in-the-loop control is maintained: AI handles message organisation, while merchants keep the final decision-making power.

3. TECHNICAL ARCHITECTURE & IMPLEMENTATION PLAN

3.1. Conceptual Tech Stack

List the technologies, frameworks, APIs, and databases you plan to use to build a functional, deployed prototype. Describe (or attach as an appendix) a simple system block diagram showing how the workshop elements connect into one architecture.

Did your project utilize any official SDC sponsor-provided GenAI API or Cloud Platform? If yes, specify which one and how it is used.

We build this prototype with the following tech stack:

1. Frontend: Simple HTML + CSS + JavaScript, for merchant dashboard, order review page and lightweight menu display page. Merchants paste raw order messages, view parsed structured orders, approve or reject entries here.
2. Backend: Python (Flask) as the web server, to handle request routing, data passing and business logic.
3. GenAI API: SDC sponsor-provided Generative AI API. It receives unstructured free-text order messages, extracts structured fields (buyer name, dish, quantity, remark, delivery address), and marks ambiguous / incomplete messages for human review.
4. Database: SQLite for local prototype storage, to store merchant menu data, order records, approval status and operation logs.
5. Hosting: Cloud hosting service for prototype deployment and public access.

System Architecture Flow:
Merchant Input (raw chat text) → Backend forwards text to GenAI API → AI returns structured parsed result → Backend saves data into SQLite → Render order list on frontend dashboard → Merchant approves/rejects orders → System generates preparation & delivery checklist.
This project uses the official SDC sponsor GenAI API. It is the core component responsible for natural language parsing and information extraction from informal group order messages

3. TECHNICAL ARCHITECTURE & IMPLEMENTATION PLAN

3.2. Design-Thinking & Deployment Strategy

Explain how your team applied design-thinking methodology to ensure this solution is technically viable, user-friendly, and ready for real deployment by Phase 2.

[Maximum 200 words]

We followed design-thinking stages: Empathise, Define, Ideate, Prototype and Test. We interviewed campus food vendors to learn their pain points of manual order sorting, then narrowed our scope to parsing unstructured chat messages instead of building a full ordering platform.

We prioritised human-in-the-loop approval to fit vendor habits and reduce AI risk. We built a minimal viable prototype first, focusing only on core parsing and order review functions. We will run iterative tests with simulated order messages, collect feedback and refine prompt rules and UI.

For deployment, we plan cloud hosting by Phase 2. We keep the system modular so merchants can configure menus easily, ensuring it is technically stable, simple for non-technical vendors to operate, and ready for real-world trial.

4. PRELIMINARY ENVIRONMENTAL & SDG IMPACT FRAMEWORK

To align with SDC's focus on meaningful global sustainability, establish your project's impact baseline before building the prototype. Use estimated data, published research, or public benchmarks where exact figures are unavailable. Provide the citation of any sources used.

The target, baseline, and KPI you provide here will be used by judges during shortlisting to assess Innovation and Creativity as well as Feasibility of build. Vague or unsupported projections may weaken your case for advancing to Phase 2.

4. PRELIMINARY ENVIRONMENTAL & SDG IMPACT FRAMEWORK

Metric Category	Your Response
Target SDG Sub-Target(s) *Identify the exact UN SDG sub-target your project addresses* *e.g., Target 11.6 "Reduce the adverse per-capita environmental impact of cities, focusing on air quality and waste management."*	SDG 8.3: Promote development-oriented policies that support productive activities, decent job creation, entrepreneurship, creativity and innovation, and encourage the formalization and growth of micro-, small- and medium-sized enterprises, including through access to financial services.
Current Problem Baseline *What is the starting condition of the problem, expressed as a number?* *e.g., "The target infrastructure currently experiences 35% energy inefficiency due to unoptimized scheduling."*	Campus micro food vendors spend roughly 30 – 45 minutes daily manually sorting unstructured chat orders, with around 15% of daily food prepared wasted due to miscounted or missed student orders.

4. PRELIMINARY ENVIRONMENTAL & SDG IMPACT FRAMEWORK

Metric Category	Your Response
Core Measurable KPI *The specific unit you will track* *e.g., % reduction in power consumption, processing-latency improvement, or kg of resource waste avoided*	Time spent daily on manual order sorting (minutes), percentage of food waste caused by order counting errors, order parsing error rate.
Phase 2 Impact Target *Your realistic, percentage-based target by the final prototype presentation* *e.g., "20% improvement in computational resource use."*	Reduce vendors ' daily order-sorting labour time by 60% and cut error-related food waste by 40% at the Phase 2 prototype trial.

5. ELIGIBILITY & READINESS CHECKLIST

Tick all boxes that apply before submitting.

- ☐ All team members have completed the mandatory GenAI, Database, and Cloud Deployment upskilling workshops.
- ☐ The project concept addresses a verified real-world problem mapped directly to one or more UN SDGs.
- ☐ The proposed solution targets a functional, deployed application rather than a theoretical business case study.
- ☐ All fields in this template have been completed within the word limits stated.

6. IP & ORIGINALITY DECLARATION

6.1. Pre-Existing / Prior-Owned Work

Does this submission include any code, design, research, or other work that any team member created or owned before the start of this hackathon?

☐ Yes ☑ No

If yes, briefly describe what was pre-existing and which team member(s) owned it:

6. IP & ORIGINALITY DECLARATION

6.2. Third-Party Code, Datasets, or Models

Does your project use any third-party code libraries, open-source frameworks, public/private datasets, or pre-trained AI models that your team did not create?

☑ Yes ☐ No

If yes, list them below (open-source use is permitted provided it is declared and properly licensed/attributed):

Name	Type (code/dataset/model)	License
DeepSeek API	Pre-trained AI Model	DeepSeek Open Platform Terms of Service

6. IP & ORIGINALITY DECLARATION

6.3. Employment-Related IP Conflicts

Is any team member currently bound by an employment, internship, or consulting agreement under which their employer could claim ownership of work they create during this hackathon?

☐ Yes ☑ No

If yes, please describe the nature of the agreement and which member it concerns:

7. DECLARATION

By submitting this proposal, the team leader confirms that the information provided is accurate to the best of the team's knowledge and that the project will be original work created during the SDC 2026 hackathon period.

[Signature]

Liu Shiwei

[Date]

24/10/2026